



# Laser Authorized User Training

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# Objectives

- Become an authorized user (able to use machines unsupervised)
- Learn normal operations procedures
- Learn safety procedures



# Requirements

- Must have passed “Intro” course
- Must be paying member
- Must agree to follow policy documents and checklist



# Policies

- You must carefully read, initial, and sign the policy document. This governs use of the machine and outlines your responsibility as operator.
- Copy provided by trainer
- Also available on GitHub, follow QR code on lasers.
- Policies
- Yes, there will be a test.



# General Provisions

- Provide your own materials
- Not under influence
- Log use and failures or safety issues
- Keep machine locked when not in use
- Follow items on checklist



# Approved Materials

- If it's not on the approved materials list, don't cut it.
- If you disagree, get a copy of the Safety Data Sheet (SDS) and appeal to board for changes to policy, we can change if presented with a well-reasoned argument.
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# Safety:

- Each type of safety condition should be handled depending on its nature.
  - Fire
  - Lens Vaporization
  - Tube Malfunction



# Fire

- Most common
- Not a problem if handled quickly
- Big problem if left unattended
- Use E-Stop
- Extinguish fire, escalating as needed
  - Turn off air assist
  - Use fire blanket
  - Use extinguisher
  - Call 911
- Log failure and notify board.





# Lens Vaporization

- Prevent by examining lens about monthly to make sure residue hasn't accumulated. If you don't want to clean it yourself, notify someone who can.
- Prevention is the most powerful tool here.
- Lenses cost ~\$40 and it's much easier, safer to replace a lens than to clean the machine.
- Keep lid closed
- Keep locked
- HAZMAT to avoid Selenium exposure
- Log failure and notify board
- Place sign on machine to warn of exposure.



# Laser tube malfunction

- Tube can crack
  - Physical shock
  - Water freezing
- High-voltage risk
- E-stop
- Log failure, notify board member
- Keep locked, closed
- Sign on machine



# Supervising others

- You may supervise general public WHO HAVE HAD INTRO TRAINING
  - We encourage this because it may help drive membership and donations
- You must watch them and stay nearby
- You are ultimately responsible for safety



# Demonstrate Safe Operation

- We will now go to the lasers and
  - Prepare them for use
    - Unlock machine
    - Exhaust vents
    - Fan power
    - Air assist
    - Chillers



# Demonstrate Safe Operation

- We will now go to the lasers and
  - Cut a small job
    - Focal length
    - Lightburn settings



# Demonstrate Safe Operation

- We will now go to the lasers and
  - Close them down
    - Lock machine
    - Exhaust vents
    - Fan power
    - Air assist
    - Chillers



# Examination

- Be prepared to answer questions about what to do in case of emergency
  - Fire
  - Lens vaporization
  - Tube malfunction



# Congratulations, you are now authorized

- Signoff
- Authorized user badge/token

